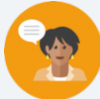


13.7 Centroids of Triangles

Lesson Introduction

 Benchmark	MA.912.GR.3.2 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.		 Learning Target	I can use coordinate geometry to identify properties of a triangle.
 Benchmark Coherence	Building On		Working Toward	
	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through a given point.		N/A	
 Essential Questions	- How do you find the centroid of a triangle by graphing the medians? - What is a median in a triangle? - What is a centroid in a triangle? - How do you find the centroid using the coordinates of the midpoints of the three sides of the triangle?			
 Lesson Narrative	Lesson 13.7 guides students to find the centroid of a triangle. The Exploration has students finding the midpoints of each side of the triangles. Then, students use the steps to construct the medians in order to find the centroid of the given triangle. In the Collaboration activity, students discover that the process for finding the centroid involves finding the averages of the x -coordinates and the y -coordinates. Students work with both strategies in the lesson.			
 Component Summary	13.7.1 Warm-Up (~5 mins)	First question to come to mind (<i>SE p. 322</i>)	13.7.3 Collaboration (10 mins)	Partner/group exercise on finding the centroid (<i>SE p. 325</i>)
	13.7.2 Exploration (20 mins)	Discovery activity on centroids of triangles (<i>SE p. 322</i>)	13.7.4 Your Turn! (10 mins)	Independent practice finding the centroid (<i>SE p. 325</i>)
	13.7.5 Wrap-Up (~5 mins)	Ticket Out the Door (<i>SE p. 326</i>)		
 Resources	Glossary		Materials	
	Key Terms: N/A		- Straightedge - Calculators (Optional): Graphing utility (Optional): Patty paper (Optional): Colored pens/pencils	
	Additional Terms: - Centroid - Midpoint - Median			
			Additional Preparation	
			If planning on using patty paper, graph paper, and/or colored pens and pencils, have these materials ready for students to use.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle with finding the centroid when decimals are involved. - Students may struggle with finding the third vertex of the triangle on question 2 in 13.7.4 Your Turn!.	- Consider allowing students to use colored pencils/pens when graphing the medians to find the centroid. - Consider allowing students to use patty paper or graph paper on certain problems.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect Lesson 13.7 is structured to help students make connections between two various strategies for finding the centroid of a triangle. Execute this lesson by implementing strategies that help students compare and connect the methods for finding the centroid of a triangle.	MA.K12.MTR.5.1: Patterns and Structure Lesson 13.7 is designed to provide students with the opportunity to make a connection between the x- and y-coordinates to find the centroid. The lesson also provides students with scaffolding necessary to discover patterns relating to finding the centroid of a triangle. Encourage students to find and look for the patterns when finding the centroid of a triangle.
 Differentiate Instruction	Consider using visuals to highlight the concept of centroids. There are several graphing utilities that students can use to discover the relationships of centroids of triangles. Another approach is to have students draw a triangle on patty paper. Students find the midpoints on each side of the triangle, and then fold each of the three segments at the midpoints to form the medians. Once the three medians are visible, have students find the point where the three medians meet to discover the centroid.	
Lesson Notes:		

13.7.1 Warm-Up: Human Hamster Wheel

Component Overview: Students examine a picture and write down the first question that comes to their mind. Students record two things that they noticed in the picture.



13.7.1 Warm-Up: Human Hamster Wheel

A human hamster wheel is shown.



Consider having students share their questions and notices with the class to hear a variety of viewpoints.

1. What is the first question that comes to your mind?
Answers will vary.
What is the diameter of this? Will the man be able to hold on if it starts rolling faster? How many of the white boxes is this made up of?
2. What are two things you notice about the human hamster wheel?
Answers will vary.
It is very large and round. The white boxes making up the wheel appear to be equal in size.

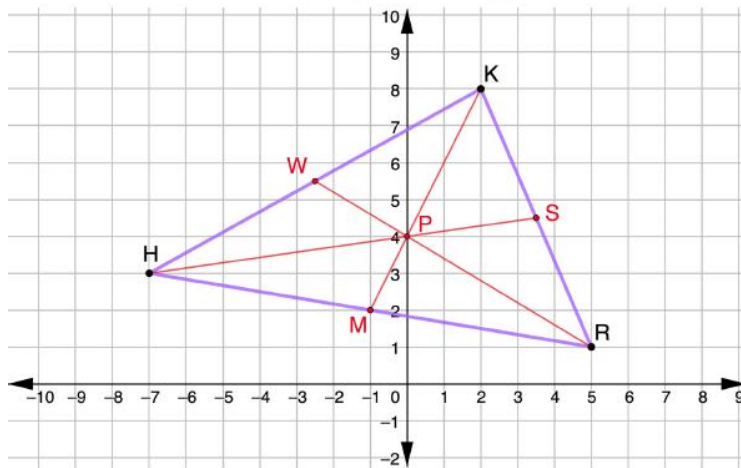
13.7.2 Exploration: Centroids of Triangles

Component Overview: This Collaboration guides students through the steps for finding the centroid of a triangle. Students start by finding midpoints of each side of the triangle, and then are guided to find the centroid using the averages of the x -coordinates and the y -coordinates.



13.7.2 Exploration: Centroids of Triangles

1. $\triangle HKR$ has vertices at $H(-7,3)$, $K(2,8)$, and $R(5,1)$.



- a. Complete the table by finding the midpoint of each line segment.

Line Segment	Work	Midpoint
$\overline{HK}$	$\left(\frac{-7+2}{2}, \frac{3+8}{2}\right)$ $\left(\frac{-5}{2}, \frac{11}{2}\right)$	$(-2.5, 5.5)$
$\overline{KR}$	$\left(\frac{2+5}{2}, \frac{8+1}{2}\right)$ $\left(\frac{7}{2}, \frac{9}{2}\right)$	$(3.5, 4.5)$
$\overline{RH}$	$\left(\frac{5-7}{2}, \frac{1+3}{2}\right)$ $\left(\frac{-2}{2}, \frac{4}{2}\right)$	$(-1, 2)$



This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. Consider where to take the opportunity to summarize and synthesize what is discovered before the next component. Allow students to revise their thinking throughout the Exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group when you summarize.

Additional scaffolding ideas and questions are provided, if needed, but this should be a primarily student-directed Exploration.



Consider having students plot the midpoints on the graph after finding the results in the table.



As students are working, ask the following questions to prompt student thinking:

- What is the midpoint of a line segment?
- Can the graphs be used to find the midpoints? If so, how?

13.7.2 Exploration: Centroids of Triangles (continued)

- b. Complete the steps to find the centroid of triangle HKR .

Shown on given triangle.

1	Graph the midpoint of line $\overline{HK}$ and label it W .
2	Using a straightedge, draw the median from W to the opposite vertex, $\angle HRK$.
3	Graph the midpoint of line $\overline{KR}$ and label it S .
4	Using a straightedge, draw the median from S to the opposite vertex, $\angle KHR$.
5	Graph the midpoint of line $\overline{RH}$ and label it M .
6	Using a straightedge, draw the median from M to the opposite vertex, $\angle HKR$.
7	Label the centroid of triangle HKR point P .

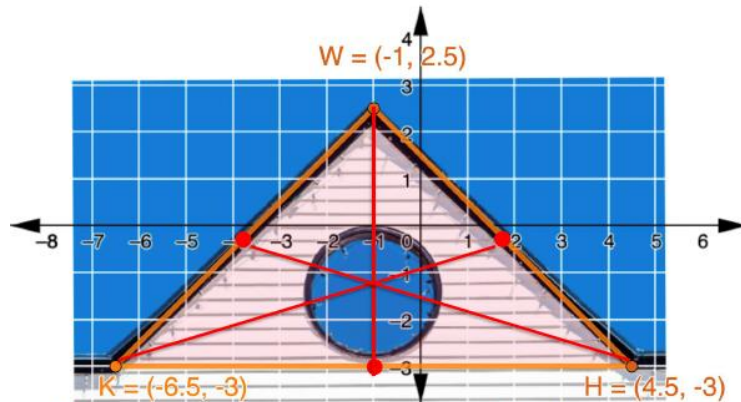
- c. What are the coordinates of the centroid of $\triangle HKR$?
 $(0, 4)$



Consider having students use three separate colors to draw the medians, so that each median is a different color. This strategy may be helpful for students to see the medians and to find the centroid.

13.7.2 Exploration: Centroids of Triangles (continued)

2. The façade of a house is shown on the coordinate grid. The façade is represented by $\triangle WHK$ with vertices $W(-1, 2.5)$, $H(4.5, -3)$, and $K(-6.5, -3)$.



Work with your partner to estimate the location of the centroid of the triangular façade.

$$\begin{aligned}\text{Midpoint of } KW &= \left(\frac{-6.5-1}{2}, \frac{-3+2.5}{2} \right) \\ &= \left(\frac{-7.5}{2}, \frac{-0.5}{2} \right) \\ &= (-3.75, -0.25)\end{aligned}$$

$$\begin{aligned}\text{Midpoint of } KH &= \left(\frac{-6.5+4.5}{2}, \frac{-3-3}{2} \right) \\ &= \left(\frac{-2}{2}, \frac{-6}{2} \right) \\ &= (-1, -3)\end{aligned}$$

$$\begin{aligned}\text{Midpoint of } WH &= \left(\frac{-1+4.5}{2}, \frac{2.5-3}{2} \right) \\ &= \left(\frac{3.5}{2}, \frac{-0.5}{2} \right) \\ &= (1.75, -0.25)\end{aligned}$$

See coordinate grid for example student work.

Answers will vary.

Centroid is approximately $(-1, -1.2)$.



Students may struggle with this problem, as there are multiple decimal coordinates in the diagram.



Consider having students use patty paper over the diagram to better see and find the centroid.

Also, consider having students make a table similar to the table in question 1a to organize their work.

13.7.2 Exploration: Centroids of Triangles (continued)

3. Use the centroids for $\triangle HKR$ and $\triangle WHK$ to complete the table.

Answers for the centroid of $\triangle WHK$ will vary because students estimated the location.

Triangle	Centroid	Average of the x -coordinates of the Vertices	Average of the y -coordinates of the Vertices
$\triangle HKR$	(0,4)	0	4
$\triangle WHK$	(-1, -1.2)	-1	-1.17

4. Work with your partner to make a conjecture about the relationships of the centroid and the average of the x and y coordinates.

The average of the x -coordinates of the vertices of the triangle gives the x -value of the centroid. The average of the y -coordinates of the vertices of the triangle gives the y -value of the centroid.

5. Ask two classmates for their conjectures. Record their conjectures and write your summary of their conjectures. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Answers will vary.

Conjecture	My Summary of Their Conjecture	Initials

6. Review your conjecture with your partner. If necessary, make any changes.

Answers will vary.



Which representation made it easier to find the centroid, the picture, or the table? Why?



Provide students with the opportunity to share their conjectures with the class, as well as with the two classmates. Look for any commonalities in the conjectures. Use the Literacy and Language Routine to have students compare their conjectures, and connect to the first method for finding the centroid.

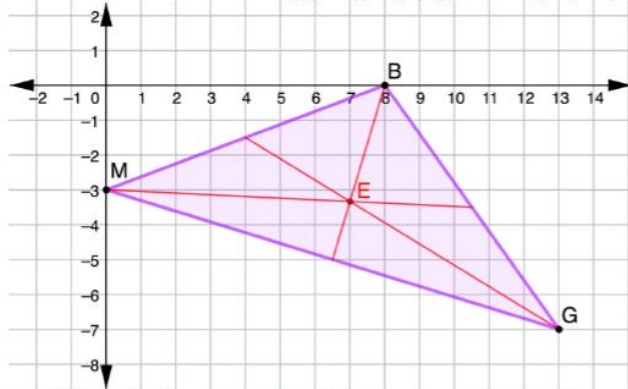
13.7.3 Collaboration: Centroids of Triangles

Component Overview: Students work together to find the centroids of triangles. Allow students to work with partners and/or small groups during this component of the lesson.



13.7.3 Collaboration: Centroids of Triangles

1. $\triangle MBG$ has vertices at $M(0, -3)$, $B(8, 0)$, and $G(13, -7)$.



- What is the average of the x -coordinates?
 7
- What is the average of the y -coordinates?
 $-\frac{10}{3}$
- What are the coordinates of the centroid?
 $(7, -\frac{10}{3})$
- Verify the coordinates of the centroid by drawing the median of each side.
Shown above.



If your students struggled creating a conjecture in 13.7.2 Exploration, then consider using 13.7.3 as a Guided Instruction or to work with a small group of students who need further support.



Encourage students to interact and discuss while completing this section of the lesson by asking clarifying questions and explaining their thought process. Have students use MLR5 to connect the patterns for the methods for finding the centroid. These questions give students the opportunity to make connections.

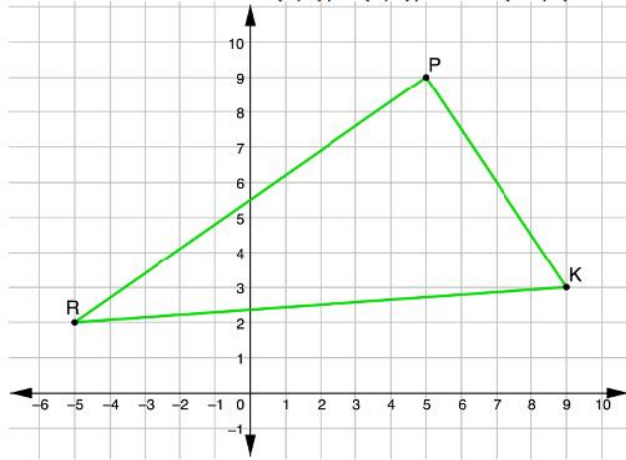
13.7.4 Your Turn!

Component Overview: Students work independently to practice finding the centroids of triangles using different representations of the given triangle.



13.7.4 Your Turn!

1. $\triangle PQR$ has vertices at $P(5,9)$, $K(9,3)$, and $R(-5,2)$.



What are the coordinates of the centroid of $\triangle PQR$?

Centroid is $(3, \frac{14}{3})$.



Monitor which strategy students are using to find the centroid. Allow students to either find the centroid by finding the average of the x and y -coordinates, or by having students find the midpoint of each side of the triangle and graphing the medians.

13.7.4 Your Turn! (continued)

2. $\triangle NHT$ has vertices at $N(-8, -3)$ and $H(3, -1)$ and a centroid at $(-2, -4)$.

What are the coordinates of vertex T ?

$$\frac{-8 + 3 + x}{3} = -2$$

$$x = -1$$

$$\frac{-3 - 1 + y}{3} = -4$$

$$y = -8$$

Vertex T is $(-1, -8)$.

3. Artemio is finding the centroid of $\triangle WYV$ with vertices at $W(-8, 5)$, $Y(7, 8)$, and $V(-2, -3)$.

His work is shown.

Artemio's Work	
$x = \frac{8 + 7 + 2}{3}$	$y = \frac{5 + 8 + 3}{3}$
$x = \frac{17}{3}$	$y = \frac{16}{3}$
Centroid $(\frac{17}{3}, \frac{16}{3})$	

- a. Write a note to Artemio explaining his mistake.
There should be negative signs in front of the 8 and 2 on the x side and in front of the 3 on the y side.
- b. What are the coordinates of the centroid of $\triangle WYV$?
 $(-1, \frac{10}{3})$



Consider opportunities to have students use a visual graphing utility to help assist with question 2.



Ask the following questions to students who are struggling with question 2:

- What is the pattern that you discovered in the lesson?
- How can you set up an equation to find the centroid?



Have a couple of students share their notes to see a variety of viewpoints.



For students who finish early, challenge them to create three coordinates and graph on a coordinate plane, or use a graphing utility. Then, have students find the centroid using either method.

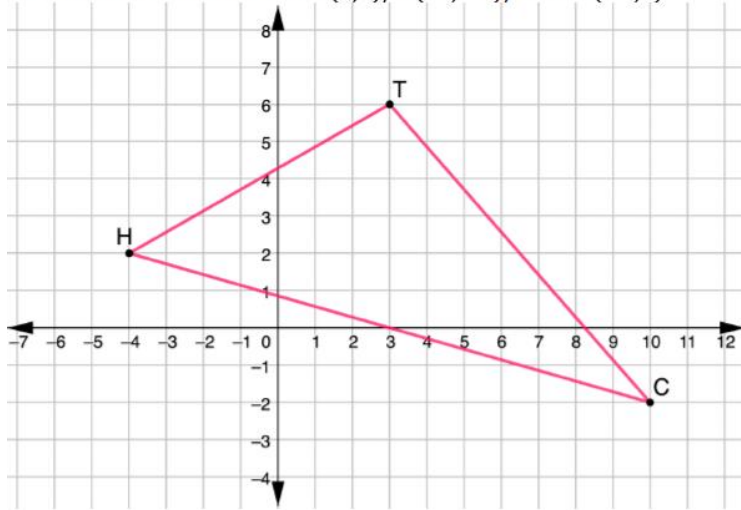
13.7.5 Wrap-Up: Ticket Out the Door

Component Overview: Students complete a question based on the learning target of the lesson.



13.7.5 Wrap-Up: Ticket Out the Door

1. $\triangle TCH$ has vertices at $T(3,6)$, $C(10,-2)$, and $H(-4,2)$.



What are the coordinates of the centroid of $\triangle TCH$?

Centroid is $(3,2)$.



Use student understanding to inform your instruction, and to determine where, during other lessons, to work with students who struggle with determining the centroid. Make a list of misconceptions, and determine which should be addressed whole-group, and which should be addressed in a small group with those who are struggling to be proficient.